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Electronic device including two-dimensional material and method of manufacturing the same

Abstract

An electronic device including a two-dimensional material is provided. The electronic device may include a substrate; a metal layer on a partial region of the substrate; a two-dimensional material layer over the metal layer and an upper surface of the substrate; and an insertion layer between the metal layer and the two-dimensional material layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0112472, filed on Aug. 25, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) The present disclosure relates to electronic devices including two-dimensional materials, and/or methods of manufacturing the electronic devices.

2. Description of Related Art

(3) Two-dimensional materials, which are materials having a two-dimensional crystal structure, may have good electrical properties and may maintain high mobility without changes in properties even when the thicknesses of the two-dimensional materials are reduced to nanoscale, and thus, the two-dimensional materials may be applied to various devices. A junction portion between a two-dimensional material and a metal may have a contact resistance due to a Schottky barrier, which may be a factor in the deterioration of electron mobility.

SUMMARY

(4) Provided are electronic devices in which a contact resistance between a two-dimensional

material and a metal is lowered, and/or methods of manufacturing the electronic devices.

(5) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

(6) According to an embodiment, an electronic device may include a substrate; a metal layer on a partial region of the substrate; a two-dimensional material layer over the metal layer and an upper surface of the substrate; and an insertion layer between the metal layer and the two-dimensional material layer.

(7) In some embodiments, the two-dimensional material layer may include a transferable two-dimensional material.

(8) In some embodiments, the two-dimensional material layer may include at least one of graphene, black phosphorus, or transition metal dichalcogenide (TMD).

(9) In some embodiments, the metal layer may include a metal having a low work function or a noble metal.

(10) In some embodiments, the insertion layer may include graphene, a-C, h-BN, or a-BN.

(11) In some embodiments, the metal layer may include a first electrode and a second electrode spaced apart from each other on the substrate. The two-dimensional material layer may cover an upper surface of the first electrode, an upper surface of the second electrode, and an upper surface of a part of the substrate between the first electrode and the second electrode.

(12) In some embodiments, the electronic device may further include a gate insulating layer on the two-dimensional material layer; and a gate electrode on the gate insulating layer. The two-dimensional material layer may be a channel layer.

(13) In some embodiments, the gate insulating layer may include at least one of a high-k oxide, a silicon oxide, or a two-dimensional insulating material capable of atomic layer deposition (ALD).

(14) In some embodiments, the metal layer may have a step with respect to a reference surface, and the two-dimensional material layer may include a first portion along the reference surface; a second portion on the metal layer; and an inclined third portion between the first portion and the second portion.

(15) In some embodiments, the two-dimensional material layer may include a transferred two-dimensional material layer, and the third portion may correspond to a suspended portion.

(16) In some embodiments, the electronic device may further include an upper metal layer on the two-dimensional material layer. The upper metal layer may correspond to the metal layer.

(17) In some embodiments, the electronic device may further include a planarization layer on the substrate. The planarization layer may expose an upper surface of the metal layer. The two-dimensional material layer may extend over the metal layer and the planarization layer.

(18) According to an embodiment, a method of manufacturing an electronic device may include forming a metal layer on a partial region of a substrate; forming an insertion layer on the metal layer; and forming a two-dimensional material layer over the metal layer and an upper surface of the substrate by a transfer process.

(19) In some embodiments, the insertion layer may be directly grown on the metal layer by a deposition process.

(20) In some embodiments, the method may further include forming a planarization layer on the substrate so planarization layer exposes an upper surface of the metal layer, and the two-dimensional material layer may extend over the metal layer and the planarization layer.

(21) The method may further forming an upper metal layer on the two-dimensional material layer, wherein the upper metal layer may correspond to the metal layer.

(22) In some embodiments, the metal layer may be formed to have a step with respect to a reference surface. The two-dimensional material layer may include a first portion along the reference surface, a second portion on the metal layer, and a third portion. The third portion may be inclined between the first portion and the second portion.

- (23) In some embodiments, the two-dimensional material layer may include a transferred two-dimensional material layer. The third portion may be a suspended portion.
- (24) In some embodiments, the method may include forming an upper metal layer on the two-dimensional material layer. The upper metal layer may correspond to the metal layer.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a schematic cross-sectional view illustrating an electronic device according to an embodiment;
- (3) FIG. 2 shows an energy barrier at a junction of a metal-two-dimensional material (semiconductor) when a two-dimensional material layer is formed, in a structure in bottom contact with a metal layer, over the metal layer protruding from an upper surface of a substrate and the upper surface of the substrate;
- (4) FIG. 3 shows, as a comparative example, an energy barrier at a junction of a metal-semiconductor in a structure when a two-dimensional material layer is in top contact with a metal layer;
- (5) FIG. 4 is a cross-sectional view schematically showing an electronic device according to another embodiment;
- (6) FIG. 5 is a cross-sectional view schematically showing an electronic device according to another embodiment;
- (7) FIG. 6 shows a transmission electron microscopy (TEM) image when a two-dimensional material layer including MoS₂ is formed over a protruding electrode by a transfer process;
- (8) FIG. 7 is a graph showing voltage-current transfer characteristics when an electronic device according to an embodiment is implemented as a field effect transistor including a bottom contact structure of a two-dimensional material layer including MoS₂;
- (9) FIG. 8 shows a comparison of mobility between a sample of an embodiment including a lower contact/upper gate and a sample of a comparative example including an upper contact/lower gate;
- (10) FIGS. 9 to 12 are cross-sectional views schematically showing electronic devices according to embodiments, and show examples implemented as field effect transistors; and
- (11) FIG. 13 is a circuit diagram of an electronic device according to an example embodiment.

DETAILED DESCRIPTION

(12) Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. For example, “at least one of A, B, and C,” “at least one of A, B, or C,” “one of A, B, C, or a combination thereof,” and “one of A, B, C, and a combination thereof,” respectively, may be construed as covering any one of the following combinations: A; B; C; A and B; A and C; B and C; and A, B, and C.”

(13) Hereinafter, example embodiments will be described with reference to the accompanying drawings. In the drawings, like reference numerals refer to like elements, and the sizes of elements may be exaggerated for clarity of illustration. The embodiments described herein are for illustrative

purposes only, and various modifications may be made therein.

(14) In the following description, when an element is referred to as being “above” or “on” another element, it may be directly on an upper, lower, left, or right side of the other element while making contact with the other element or may be above an upper, lower, left, or right side of the other element without making contact with the other element. The terms of a singular form may include plural forms unless otherwise mentioned. It will be further understood that the terms “comprises” and/or “comprising” used herein specify the presence of stated features or elements, but do not preclude the presence or addition of one or more other features or elements.

(15) An element referred to with the definite article or a demonstrative pronoun may be construed as the element or the elements even though it has a singular form. Operations of a method may be performed in an appropriate order unless explicitly described in terms of order or described to the contrary, and are not limited to the stated order thereof.

(16) In the present disclosure, terms such as “unit” or “module” may be used to denote a unit that has at least one function or operation and is implemented with hardware, software, or a combination of hardware and software.

(17) Furthermore, line connections or connection members between elements depicted in the drawings represent functional connections and/or physical or circuit connections by way of example, and in actual applications, they may be replaced or embodied with various additional functional connections, physical connections, or circuit connections.

(18) Examples or example terms are just used herein to describe technical ideas and should not be considered for purposes of limitation unless defined by the claims.

(19) An electronic device according to an embodiment includes a metal layer, a two-dimensional material layer provided on the metal layer, and an insertion layer between the metal layer and the two-dimensional material layer so that the two-dimensional material layer is formed in a structure in bottom contact with the metal layer. When first and second electrodes spaced apart from each other by the metal layer are formed, and the two-dimensional material layer is provided to cover upper surfaces of the first and second electrodes and is applied as a channel layer, the electronic device according to an embodiment may be implemented as a transistor, for example, a field effect transistor, having the structure in which the two-dimensional material layer is in bottom contact with the metal layer.

(20) In the electronic device according to an embodiment, a Fermi level pinning phenomenon that generates an energy barrier on a semiconductor surface at a junction of a metal-semiconductor may be limited and/or prevented, by forming the two-dimensional material layer in the structure in bottom contact with the metal layer.

(21) For example, due to interfacial states caused by combining with the two-dimensional material during metal deposition, there is a limitation in that the Fermi level pinning phenomenon occurs, and a height of the Schottky barrier that determines the contact resistance does not change, even if a work function is well controlled. Meanwhile, in the electronic device according to an embodiment, a reaction between the two-dimensional material and a contact metal may be suppressed, by forming the two-dimensional material layer in the structure in bottom contact with the metal layer, and thus a Fermi level pinning effect may be reduced and/or minimized. For example, in the electronic device according to an embodiment, a metal-two-dimensional material interface without chemical bonding may be formed by transferring a two-dimensional material onto a patterned metal electrode, and thus the Fermi level pinning effect may be limited and/or minimized.

(22) In addition, in the electronic device according to an embodiment, by providing the insertion layer between the metal layer and the two-dimensional material layer, a barrier may be lowered by controlling the work function, and the contact resistance may be further lowered by limiting and/or preventing oxidation of the metal layer. The oxidation of the metal layer may be reduced and/or minimized by depositing the insertion layer directly on a surface of the metal layer through in-situ

deposition after deposition of the metal layer.

(23) In addition, in the electronic device according to an embodiment, the oxidation of the metal layer may be limited and/or prevented, by providing the insertion layer between the metal layer and the two-dimensional material layer, and thus the metal layer may have more options for material selection. That is, various metals, such as a noble metal and a metal easily oxidized, may be used as a metal layer material.

(24) In addition, in the electronic device according to an embodiment, oxidation of the interface of the metal layer may be limited and/or prevented, by providing the insertion layer between the metal layer and the two-dimensional material layer, so that the two-dimensional material layer may be transferred in an external environment, and also the two-dimensional material layer may be transferred in an anti-oxidation environment such as a glove box connected to a vacuum facility.

(25) FIG. 1 is a schematic cross-sectional view illustrating an electronic device 1 according to an embodiment.

(26) Referring to FIG. 1, the electronic device 1 may include a metal layer 20 formed in a partial region on a substrate 11, a two-dimensional material layer 50 provided over the metal layer 20 and an upper surface of the substrate 11, and an insertion layer 30 between the metal layer 20 and the two-dimensional material layer 50. The insertion layer 30 may be an anti-oxidation layer that limits and/or prevents oxidation of the metal layer 20.

(27) The substrate 11 may be an insulating substrate, or a semiconductor substrate having an insulating layer formed on its surface. The substrate 11 may include, for example, a non-metal material such as silicon oxide, aluminum oxide, hafnium oxide, etc. The semiconductor substrate may include, for example, Si, Ge, SiGe, or a group III-V semiconductor material.

(28) The metal layer 20 may include a metal material having excellent electrical conductivity. The metal layer 20 may include, for example, a noble metal or a metal having a low work function. The metal layer 20 may include, for example, the noble metal such as Au, Ag, Ru, Pd, or Pt. In addition, the metal layer 20 may include the metal having the low work function, such as Ti, Al, In, or Sc. In addition, the metal layer 20 may include various metals capable of being processed in a glove box or in a vacuum. Oxidation of the metal layer 20 may be limited and/or prevented, by providing the insertion layer 30 on the metal layer 20, so that the metal layer 20 may have more options for material selection, and various types of metals may be applied to the metal layer 20. That is, not only the noble metal, but also the metal having the low work function, such as Ti, Al, In, Sc, etc., having relatively strong oxidizing properties may be applied to the metal layer 20.

(29) The insertion layer 30 may include a material that maintains or lowers the work function of the metal layer 20 formed thereunder. The insertion layer 30 may reduce a barrier by controlling the work function, and may lower a contact resistance by limiting and/or preventing the oxidation of the metal layer 20.

(30) The insertion layer 30 may include, for example, any one material selected from graphene, amorphous carbon (a-C), h-BN, and a-BN, or may include a variety of other barrier materials. The insertion layer 30 may have a thickness equal to or less than about 5 nm, for example, equal to or less than 3 nm.

(31) The insertion layer 30 may be directly grown on the metal layer 20 by a deposition process. The insertion layer 30 may be directly deposited on the surface of the metal layer 20, for example, through an in-situ deposition after deposition of the metal layer 20. As described above, when the insertion layer 30 is formed through the in-situ deposition after the deposition of the metal layer 20, the oxidation of the metal layer 20 may be reduced and/or minimized. For example, when the insertion layer 30 is formed by directly depositing, for example, graphene on the surface of the metal layer 20 through the in-situ deposition after the deposition of the metal layer 20, the oxidation of the metal layer 20 may be reduced and/or minimized.

(32) Here, when the insertion layer 30 is formed through the in-situ deposition after the deposition of the metal layer 20, the oxidation of the metal layer 20 may be reduced and/or minimized, but the

embodiment is not limited thereto. For example, when the metal layer **20** includes the noble metal, the insertion layer **30** may be formed through the in-situ deposition after the deposition of the metal layer **20**, or through other deposition methods than the in-situ deposition.

(33) The two-dimensional material layer **50** may include at least one of all types of transferable two-dimensional materials. In addition, the two-dimensional material layer **50** may include a transferable two-dimensional material exhibiting a semiconductor property. For example, the two-dimensional material layer **50** may include at least one of, for example, graphene, black phosphorous, or transition metal dichalcogenide (TMD).

(34) Graphene is a material having a hexagonal honeycomb structure in which carbon atoms are two-dimensionally bonded. Compared to silicon (Si), graphene has advantage of high electrical mobility and excellent thermal properties, chemical stability, and a large surface area. Black phosphorus is a material in which black phosphorous atoms are two-dimensionally bonded. The TMD may include a compound of a transition metal and a chalcogen element. For example, the TMD may include MoS.sub.2, MoSe.sub.2, MoTe.sub.2, WS.sub.2, WSe.sub.2, WTe.sub.2, ZrS.sub.2, ZrSe.sub.2, HfS.sub.2, HfSe.sub.2, NbSe.sub.2, ReSe.sub.2, etc. but is not limited thereto.

(35) In the electronic device **1** according to an embodiment, for example, the metal layer **20** may be formed on a partial region on the substrate **11**, the insertion layer **30** may be formed on the metal layer **20**, and the two-dimensional material layer **50** may be formed over the metal layer **20** and the upper surface of the substrate **11** by a transfer method.

(36) As described above, in the electronic device **1** according to an embodiment, since the two-dimensional material layer **50** is formed in a structure in bottom contact with the metal layer **20**, the two-dimensional material layer **50** may be transferred onto the patterned metal layer **20** to form a junction of a metal-two-dimensional material (semiconductor) without chemical bonding, and thus a Fermi level pinning effect may be reduced and/or minimized. That is, in the electronic device **1** according to an embodiment, the Fermi level pinning phenomenon that generates an energy barrier on a surface of the two-dimensional material layer **50** at the junction of the metal-two-dimensional material (semiconductor) may be limited and/or prevented.

(37) In addition, in the electronic device **1** according to the embodiment, by providing the insertion layer **30** between the two-dimensional material layer **50** and the metal layer **20**, the barrier may be lowered by controlling the work function, and the contact resistance may be lowered by limiting and/or preventing the oxidation of the metal layer **20**. In addition, by providing the insertion layer **30**, the oxidation of the metal layer **20** may be limited and/or prevented and a range of material selection for the metal layer **20** may be widened. The insertion layer **30** is deposited directly on the surface of the metal layer **20** through the in-situ deposition after the deposition of the metal layer **20**, thereby minimizing the oxidation of the metal layer **20**.

(38) Therefore, in the electronic device **1** according to an embodiment, a contact barrier, for example, a Schottky barrier, at the junction of the metal-two-dimensional material (semiconductor) may be lowered.

(39) Meanwhile, in the electronic device **1** according to an embodiment, the metal layer **20** may protrude from a reference surface, as shown in FIG. 1, whereby the metal layer **20** may have a step with respect to the reference surface. The reference surface may correspond to the upper surface of the substrate **11**. When an insulating layer is provided on the substrate **11**, the upper surface of the substrate **11** may correspond to an upper surface of the insulating layer.

(40) As shown in FIG. 1, when the metal layer **20** protrudes from the reference surface of the substrate **11**, the metal layer **20** may have a thickness, for example, equal to or greater than about 50 nm (e.g., in a range of 50 nm to 100 nm, but not limited thereto), so that a strain may be applied to the two-dimensional material layer **50** provided to cover the metal layer **20** and the upper surface of the substrate **11** by a transfer process.

(41) As described above, when the metal layer **20** has the step with respect to the reference surface,

the two-dimensional material layer **50** may include a first portion positioned along the reference surface, for example, the upper surface of the substrate **11**, a second portion positioned on the metal layer **20**, and a third inclined portion between the first and second portions. When the two-dimensional material layer **50** is formed by the transfer process, the third portion may be suspended.

(42) As described above, when the metal layer **20** protrudes from the upper surface of the substrate **11**, and the two-dimensional material layer **50** is formed over the metal layer **20** and the upper surface of the substrate **11** by the transfer process, the two-dimensional material layer **50** is suspended at a step portion due to a step between the metal layer **20** and the upper surface of the substrate **11**, whereby the strain may be additionally applied to the two-dimensional material layer **50**. The contact barrier at the junction of the metal-two-dimensional material (semiconductor) may be further lowered by the additionally applied strain.

(43) FIG. 2 shows an energy barrier at a junction of a metal-two-dimensional material (semiconductor) when the two-dimensional material layer **50** is formed, in a structure in bottom contact with the metal layer **20**, over the metal layer **20** protruding from the upper surface of the substrate **11** and the upper surface of the substrate **11**. In FIG. 2, CB denotes a conduction band, VB denotes a valence band, E_f denotes a Fermi level, and ϕ denotes a Schottky barrier.

(44) As may be seen from FIG. 2, when the two-dimensional material layer **50** is formed in the structure in bottom contact with the metal layer **20**, and a strain is additionally applied to the two-dimensional material layer **50** suspended by a step between the metal layer **20** and the upper surface of the substrate **11**, the energy barrier, that is, the Schottky barrier, generated on a semiconductor surface at the junction of the metal-semiconductor may be further lowered by the strain applied to the two-dimensional material layer **50**, and a bandgap may also be narrowed by the strain.

(45) That is, in the electronic device **1** according to an embodiment, Fermi level de-pinning may be induced, by forming the two-dimensional material layer **50** in the structure in bottom contact with the metal layer **20**, and thus contact barrier lowering may be obtained. In addition, a tensile strain is additionally applied to the two-dimensional material layer **50** suspended by a step between the metal layer **20** and the upper surface of the substrate **11**, and thus band gap narrowing may be obtained, thereby further lowering the Schottky barrier.

(46) In addition, in the electronic device **1** according to an embodiment, the two-dimensional material layer **50** may be formed by a transfer process, by forming the two-dimensional material layer **50** in the structure in bottom contact with the metal layer **20**, and thus, a large-area integration process is possible.

(47) To the contrary, as in the comparative example of FIG. 3, in a structure in which the two-dimensional material layer **50** is in top contact with the metal layer **20**, when the metal layer **20** is formed by the transfer process so that the two-dimensional material layer **50** is not damaged, the large-area integration process is impossible.

(48) FIG. 3 shows, as a comparative example, an energy barrier at a junction of a metal-semiconductor in a structure when a two-dimensional material layer **50'** is in top contact with a metal layer **20'**. In FIG. 3, CB denotes a conduction band, VB denotes a valence band, E_f denotes a Fermi level, and ϕ denotes a Schottky barrier.

(49) As shown in FIG. 3, it may be seen that when the two-dimensional material layer **50'** has a structure in top contact with the metal layer **20'**, a large energy barrier, that is, a large Schottky barrier, is generated by a Fermi level pinning phenomenon that occurs at the junction of the metal-semiconductor.

(50) Meanwhile, the electronic device **1** according to an embodiment including the metal layer **20** protruding from the upper surface of the substrate **11** has been described above as an example, but the embodiment is not limited thereto. As shown in FIG. 4, the metal layer **20** may be formed in a structure that does not protrude from the upper surface of the substrate **11**.

(51) FIG. 4 is a cross-sectional view schematically showing an electronic device **10** according to another embodiment. Compared with FIG. 1, FIG. 4 shows the two-dimensional material layer **50** provided flatly on the metal layer **20** and the substrate **11**.

(52) Referring to FIG. 4, the electronic device **10** according to an embodiment may include the metal layer **20** formed in a partial region on the substrate **11**, a planarization layer **15** formed on the substrate **11** to bury the metal layer **20** so that an upper surface of the metal layer **20** is exposed, the two-dimensional material layer **50** provided flatly over the metal layer **20** and the planarization layer **15**, and the insertion layer **30** between the metal layer **20** and the two-dimensional material layer **50**. The same reference numeral as in FIG. 1 denote substantially the same configuration, and thus a repetitive description thereof is omitted herein.

(53) The planarization layer **15** may include an insulating material, for example, silicon oxide. As shown in FIG. 4, the two-dimensional material layer **50** may be formed in a structure in bottom contact with the metal layer **20** and without additional strain application, by applying the planarization layer **15** burying the metal layer **20** so that the upper surface of the metal layer **20** is exposed, and forming flatly the two-dimensional material layer **50** over the metal layer **20** and the planarization layer **15** by a transfer process.

(54) As shown in FIG. 4, when the planarization layer **15** burying the metal layer **20** is provided so that the upper surface of the metal layer **20** is exposed, the two-dimensional material layer **50** may be formed in the structure in bottom contact with the metal layer **20**, even if a two-dimensional material that is easily broken by strain during the transfer process is applied as the two-dimensional material layer **50**.

(55) As shown in FIG. 4, when the planarization layer **15** burying the metal layer **20** so that the upper surface of the metal layer **20** is exposed is applied, the two-dimensional material layer **50** may be formed as a flat bottom contact structure.

(56) A structure in which the planarization layer **15** burying the metal layer **20** so that the upper surface of the metal layer **20** is exposed is applied may be formed in the following manner according to a material forming the metal layer **20**.

(57) For example, when a metal capable of a planarization process, for example, a chemical mechanical polishing (CMP) process, such as a slurry, is used, in a state in which the planarization layer **15** is formed on the substrate **11** and a trench is previously deeply formed in a region where the metal layer **20** is to be formed, a thick metal is deposited on the entire surface of the trench and then planarized by the planarization process, for example, the CMP process, and thus the structure in which the planarization layer **15** burying the metal layer **20** is applied so that the upper surface of the metal layer **20** is exposed without an air gap may be formed.

(58) As another example, when a metal incapable of the planarization process, for example, the CMP process, such as a noble metal, is used, the planarization layer **15** is formed on the substrate **11**, and a trench is shallowly formed to have various inclinations in a region in which the metal layer **20** is to be formed, and thus the structure in which the planarization layer **15** burying the metal layer **20** so that the upper surface of the metal layer **20** is exposed is applied may be formed through a lift-off process in which the metal is deposited only up to the height of the trench.

(59) The two-dimensional material layer **50** may be formed in a flat bottom contact structure, by forming the insertion layer **30** on the metal layer **20** of the planarized structure as formed above, and transferring the two-dimensional material layer **50** on the planarized structure.

(60) The electronic devices **1** and **10** having the above-described structure may be implemented as field effect transistors of various types, as shown in FIGS. 5 and 9 to 11.

(61) According to electronic devices **100**, **200**, **300**, **400**, and **500** according to embodiments shown in FIGS. 5 and 9 to 12, a field effect transistor having a low contact barrier between a two-dimensional material and a metal may be implemented, by introducing a lower electrode structure in which the insertion layer **30** is present.

(62) FIG. 5 is a cross-sectional view schematically showing the electronic device **100** according to

another embodiment, and shows an example implemented as a field effect transistor.

(63) Referring to FIG. 5, the electronic device **100** according to an embodiment includes first and second electrodes **121** and **125** provided on a substrate **110** to be spaced apart from each other, a two-dimensional material layer **150**, and insertion layers **131** and **135** between the first and second electrodes **121** and **125** and the two-dimensional material layer **150**. The two-dimensional material layer **150** may be provided to cover upper surfaces of the first and second electrodes **121** and **125** and an upper surface of the substrate **110** between the first and second electrodes **121** and **125**. The electronic device **100** according to an embodiment may further include a gate insulating layer **160** formed on the two-dimensional material layer **150**. In addition, the electronic device **100** according to an embodiment may further include a gate electrode **170** on the gate insulating layer **160**. In addition, the electronic device **100** according to an embodiment may further include a lower gate electrode **190**. Here, the electronic device **100** according to an embodiment may include only the gate electrode **170** without the lower gate electrode **190**. Also, the electronic device **100** according to an embodiment may include only the lower gate electrode **190** without the gate insulating layer **160** and the gate electrode **170**.

(64) The substrate **110** may be an insulating substrate, or a substrate having an insulating layer **111** formed thereon. For example, the substrate **110** may include the insulating layer **111** including a non-metal material such as silicon oxide, aluminum oxide, or hafnium oxide. Also, the substrate **110** may include a semiconductor substrate. In this regard, the semiconductor substrate may include, for example, Si, Ge, SiGe, or an III-V group semiconductor material. For example, the substrate **110** may be a semiconductor substrate having the insulating layer **111** formed on the surface thereof. The substrate **110** may be, for example, a silicon substrate having silicon oxide formed on the surface thereof, but is not limited thereto.

(65) FIG. 5 shows an example in which the substrate **110** includes the insulating layer **111** and the lower gate electrode **190** is buried in the insulating layer **111** so that the lower gate electrode **190** corresponds to a channel region.

(66) The first and second electrodes **121** and **125** may include a metal material having excellent electrical conductivity, and may be formed on the substrate **110** to be spaced apart from each other. Any one of the first and second electrodes **121** and **125** may be a source electrode, and the other may be a drain electrode.

(67) The first and second electrodes **121** and **125** may be formed of the metal layer **20** described above. The first and second electrodes **121** and **125** may include, for example, a noble metal or a metal having a low work function. The first and second electrodes **121** and **125** may include, for example, the noble metal such as Au, Ag, Ru, Pd, or Pt. In addition, the first and second electrodes **121** and **125** may include the metal having the low work function, such as Ti, Al, In, or Sc. In addition, the first and second electrodes **121** and **125** may include various metals capable of being processed in a glove box or in a vacuum. Oxidation of the first and second electrodes **121** and **125** may be limited and/or prevented, by providing the insertion layers **131** and **135** on the first and second electrodes **121** and **125**, respectively, so that the first and second electrodes **121** and **125** may have more options for selection of metal materials, and various types of metal may be applied to the first and second electrodes **121** and **125**. That is, not only the noble metal, but also the metal having the low work function, such as Ti, Al, In, Sc, etc., having relatively strong oxidizing properties may be applied to the first and second electrodes **121** and **125**.

(68) As shown in FIG. 5, the first and second electrodes **121** and **125** may be spaced apart from each other on the substrate **110** to have a step with respect to a reference surface. The reference surface may correspond to the upper surface of the substrate **110**. That is, the first and second electrodes **121** and **125** may be formed to be spaced apart from each other on the substrate **110** and protrude from the surface of the substrate **110**.

(69) As shown in FIG. 5, when the first and second electrodes **121** and **125** are formed to be spaced apart from each other on the substrate **110** and protrude from the surface of the substrate **110**, the

first and second electrodes **121** and **125** may be formed to have a thickness, for example, equal to or greater than about 50 nm so that a strain may be applied to the two-dimensional material layer **150** provided to cover the upper surfaces of the first and second electrodes **121** and **125** and the upper surface of the substrate **110** between the first and second electrodes **121** and **125**.

(70) The insertion layers **131** and **135** may be formed on the first and second electrodes **121** and **125** respectively. The insertion layers **131** and **135** may be anti-oxidation layers that limit and/or prevent oxidation of the surfaces of the first and second electrodes **121** and **125** respectively. As shown in FIG. 5, when the first and second electrodes **121** and **125** protrude to be spaced apart from each other on the substrate **110**, the insertion layers **131** and **135** may be respectively formed on the first and second electrodes **121** and **125** that protrude.

(71) The insertion layers **131** and **135** may be formed of a material that maintains or lowers the work function of the first and second electrodes **121** and **125** formed thereunder. The insertion layers **131** and **135** may reduce a barrier by controlling the work function, and may lower a contact resistance by limiting and/or preventing the oxidation of surfaces of the first and second electrodes **121** and **125**.

(72) The insertion layers **131** and **135** may include, for example, any one material selected from graphene, amorphous carbon (a-C), h-BN, and a-BN, or may include various other barrier materials. The insertion layers **131** and **135** may be formed to have a thickness of about 5 nm or less, for example, about 3 nm or less. The insertion layers **131** and **135** may be directly grown on the first and second electrodes **121** and **125** respectively by a deposition process. The insertion layers **131** and **135** may be directly deposited on the surfaces of the first and second electrodes **121** and **125**, respectively, for example, through an in-situ deposition after deposition of the first and second electrodes **121** and **125**. As described above, when the insertion layers **131** and **135** are formed through the in-situ deposition after the deposition of the first and second electrodes **121** and **125**, the oxidation of surfaces of the first and second electrodes **121** and **125** may be reduced and/or minimized. For example, when the insertion layers **131** and **135** are formed by directly depositing, for example, graphene on the surfaces of the first and second electrodes **121** and **125** through the in-situ deposition after the deposition of the first and second electrodes **121** and **125**, the oxidation of the surfaces of the first and second electrodes **121** and **125** may be reduced and/or minimized. Here, when the insertion layers **131** and **135** are formed through the in-situ deposition after the deposition of the first and second electrodes **121** and **125**, the oxidation of the first and second electrodes **121** and **125** may be reduced and/or minimized, but the embodiment is not limited thereto. For example, when the first and second electrodes **121** and **125** are formed of a noble metal, the insertion layers **131** and **135** may be formed through the in-situ deposition after the deposition of the first and second electrodes **121** and **125**, or through other deposition methods than the in-situ deposition. For example, the insertion layers **131** and **135** may be formed by transferring a single layer or several layers of graphene (e.g., 3 or less, 5 or less, or 10 or less) onto the surfaces of the first and second electrodes **121** and **125**.

(73) The two-dimensional material layer **150** may form a channel layer. The two-dimensional material layer **150** may be formed to cover the upper surfaces of the first and second electrodes **121** and **125** and the upper surface of the substrate **110** between the first and second electrodes **121** and **125** by the transfer process.

(74) For example, the first and second electrodes **121** and **125** may be formed on the substrate **110** to be spaced apart from each other, by depositing a metal layer forming the first and second electrodes **121** and **125** on the substrate **110** and patterning the metal layer. The insertion layers **131** and **135** may be respectively formed on the first and second electrodes **121** and **125**, and the two-dimensional material layer **150** may be formed over the upper surfaces of the first and second electrodes **121** and **125** and the upper surface of the substrate **110** between the first and second electrodes **121** and **125** by the transfer process.

(75) For another example, the first and second electrodes **121** and **125** respectively having the

insertion layers **131** and **135** on their surfaces to be spaced apart from each other may be formed on the substrate **110**, by depositing a metal layer on the substrate **110**, and, before patterning, forming the insertion layers **131** and **135** on positions of the metal layer where the first and second electrodes **121** and **125** are to be formed, and then patterning the metal layer. In addition, the two-dimensional material layer **150** may be formed over the upper surfaces of the first and second electrodes **121** and **125** and the upper surface of the substrate **110** between the first and second electrodes **121** and **125** by the transfer process.

(76) The two-dimensional material layer **150** may be a channel layer, and a portion positioned between the first and second electrodes **121** and **125** may correspond to a channel region. For example, as shown in FIG. 5, when the first and second electrodes **121** and **125** are formed to have a step with respect to a reference surface, that is, protrude from the surface of the substrate **110**, the two-dimensional material layer **150** may include a first portion **151** positioned along the reference surface, a second portion **155** positioned on the first and second electrodes **121** and **125** with the insertion layers **131** and **135** interposed therebetween, and an inclined third portion **153** between the first and second portions **151** and **155**. The first portion **151** positioned along the reference surface, for example, the surface of the substrate **110**, may correspond to the channel region. Since the first and second electrodes **121** and **125** protrude from the substrate **110**, the third portion **153** may be suspended.

(77) Since the first and second electrodes **121** and **125** protrude from the substrate **110**, when the two-dimensional material layer **150** is formed over the upper surfaces of the first and second electrodes **121** and **125** and the upper surface of the substrate **110** between the first and second electrodes **121** and **125** by the transfer process, a strain is additionally applied to the third portion **153**, that is, the suspended portion, and, as described with reference to FIG. 2, an energy barrier, that is, the Schottky barrier, generated at a junction of a metal- two-dimensional material (semiconductor) may be lowered by the strain, and a bandgap may also be narrowed.

(78) That is, Fermi level de-pinning may be induced, by forming the two-dimensional material layer **150** in the structure in bottom contact with the first and second electrodes **121** and **125**, and thus contact barrier lowering may be obtained. In addition, a tensile strain is additionally applied to the two-dimensional material layer **150** suspended by a step between the first and second electrodes **121** and **125** and the upper surface of the substrate **11**, and thus band gap narrowing may be obtained, thereby further lowering the Schottky barrier. In addition, the two-dimensional material layer **150** may be formed by the transfer process, by forming the two-dimensional material layer **50** in the structure in bottom contact with the metal layer, and thus, a large-area integration process is possible.

(79) In the electronic device **100** according to an embodiment, the two-dimensional material layer **150** may include at least one transferable two-dimensional material selected from all types of transferable two-dimensional materials. In addition, the two-dimensional material layer **150** may include a transferable two-dimensional material exhibiting a characteristic of a semiconductor material. As described above, the two-dimensional material layer **150** may include at least one of, for example, graphene, black phosphorous, or TMD. The TMD may include a compound of a transition metal and a chalcogen element. For example, the TMD may include MoS.sub.2, MoSe.sub.2, MoTe.sub.2, WS.sub.2, WSe.sub.2, WTe.sub.2, ZrS.sub.2, ZrSe.sub.2, HfS.sub.2, HfSe.sub.2, NbSe.sub.2, ReSe.sub.2, etc. but is not limited thereto.

(80) The gate insulating layer **160** may be positioned between the two-dimensional material layer **150** and the gate electrode **170**, and the gate electrode **170** may be formed on the gate insulating layer **160** so that the gate electrode **170** corresponds to the channel region of the two-dimensional material layer **150**.

(81) The gate insulating layer **160** may include at least one of a high-k oxide, silicon oxide, and a two-dimensional insulating material capable of atomic layer deposition (ALD). The gate insulating layer **160** may include, for example, aluminum oxide, hafnium oxide, zirconium hafnium oxide,

lanthanum oxide, etc. as the high-k oxide. The gate insulating layer **160** may include, for example, h-BN as the two-dimensional insulating material. However, the embodiment is not limited thereto. (82) The gate electrode **170** may include a metal material or a conductive oxide. Here, the metal material may include, for example, at least one selected from the group consisting of Au, Ti, TiN, TaN, W, Mo, WN, Pt, and Ni. The conductive oxide may include, for example, indium tin oxide (ITO), indium zinc oxide (IZO), etc. The gate electrode **170** may include the same material as a source electrode and a drain electrode.

(83) Meanwhile, FIG. 5 shows the electronic device **100** according to an embodiment further including the lower gate electrode **190**, but this is merely an example, and the electronic device **100** according to an embodiment may include only the gate electrode **170** without the lower gate electrode **190**. For another example, the electronic device **100** according to an embodiment may include only the lower gate electrode **190** without the gate electrode **170**.

(84) The lower gate electrode **190**, like the gate electrode **170**, may include a metal material or a conductive oxide. Here, the metal material may include, for example, at least one selected from the group consisting of Au, Ti, TiN, TaN, W, Mo, WN, Pt, and Ni. The conductive oxide may include, for example, ITO, IZO, etc. The lower gate electrode **190** may include the same material as the source electrode and the drain electrode. Also, the lower gate electrode **190** may include the same material as or a different material from the gate electrode **170**.

(85) The electronic device **100** according to an embodiment may be manufactured, for example, as follows. The first and second electrodes **121** and **125** spaced apart from each other and protruding from the substrate **110** may be formed, by forming a metal layer on the substrate **110** and patterning the metal layer. In addition, the insertion layers **131** and **135** may be respectively formed on the first and second electrodes **121** and **125** through, for example, an in-situ process. Next, a two-dimensional material layer may be transferred and formed in a contact structure without chemical bonding with the first and second electrodes **121** and **125**. In this regard, the two-dimensional material layer may be suspended by the first and second electrodes **121** and **125** protruding from the substrate **110** during a transfer process, and an additional strain may be applied thereto. After the two-dimensional material layer is formed by the transfer process as described above, an oxide used as the gate insulating layer **160**, for example, a high-k oxide, may be deposited by, for example, an ALD process. In addition, the gate electrode **170** including a metal material may be formed on the gate insulating layer **160**.

(86) FIG. 6 shows a transmission electron microscopy (TEM) image when the two-dimensional material layer **150** including MoS.sub.2 is formed over a protruding electrode by a transfer process. Referring to FIG. 6, when the two-dimensional material layer **150** including MoS.sub.2 is formed over the protruding electrode by the transfer process, it may be observed that the two-dimensional material layer **150** including MoS.sub.2 rises well on the electrode without being damaged and is well suspended without breaking even by a strain applied during the transfer process.

(87) FIG. 7 is a graph showing voltage-current transfer characteristics when the electronic device **100** according to an embodiment is implemented as a field effect transistor including a bottom contact structure of the two-dimensional material layer **150** including MoS.sub.2. FIG. 8 shows a comparison of mobility between a sample of an embodiment including a lower contact/upper gate and a sample of a comparative example including an upper contact/lower gate.

(88) As may be seen from FIGS. 7 and 8, an on/off ratio is up to 10^{10} and the mobility is up to about $30 \text{ cm}^2/\text{Vs}$, making it possible to implement the field effect transistor including the lower contact/upper gate having excellent characteristics.

(89) In the electronic device **100** according to an embodiment, since the two-dimensional material layer **150** is formed in the structure in bottom contact with the first and second electrodes **121** and **125**, a junction of a metal-two-dimensional material (semiconductor) without chemical bonding may be formed by transferring the two-dimensional material layer **150** onto the patterned first and second electrodes **121** and **125**, and accordingly, the Fermi level pinning phenomenon may be

reduced and/or minimized. That is, in the electronic device **100** according to an embodiment, the Fermi level pinning phenomenon that generates an energy barrier on a surface of the two-dimensional material layer **150** at the junction of the metal-two-dimensional material (semiconductor) may be limited and/or prevented.

(90) In addition, in the electronic device **100** according to an embodiment, by providing the insertion layers **131** and **135** between the two-dimensional material layer **150** and the first and second electrodes **121** and **125**, a barrier may be lowered by controlling a work function, and a contact resistance may be lowered by limiting and/or preventing oxidation of surfaces of the first and second electrodes **121** and **125**.

(91) Therefore, in the electronic device **100** according to an embodiment, a contact barrier, for example, a Schottky barrier, at a junction of the first and second electrodes **121** and **125** and the two-dimensional material layer **150** may be lowered.

(92) In addition, in the electronic device **100** according to an embodiment, by forming the first and second electrodes **121** and **125** to have a step with respect to a reference surface, for example, the surface of the substrate **110**, when the two-dimensional material layer **50** is formed over the first and second electrodes **121** and **125** and the upper surface of the substrate **11** by the transfer process, the two-dimensional material layer **50** is suspended at a step portion, whereby the strain may be additionally applied to the suspended portion of the two-dimensional material layer **50**. By the additionally applied strain, as described with reference to FIG. 2, the energy barrier, that is, the Schottky barrier, generated on the surface of the two-dimensional material layer **150** at the junction of the metal-two-dimensional material (semiconductor) may be further lowered, and the bandgap may also be narrowed, and thus the contact barrier at the junction of the first and second electrodes **121** and **125** and the two-dimensional material layer **150** may be additionally lowered.

(93) FIG. 9 is a cross-sectional view schematically showing the electronic device **200** according to an embodiment, and shows an example implemented as a field effect transistor. The electronic device **200** shown in FIG. 9 is different from FIG. 5 in that the electronic device **200** further includes first and second upper electrodes **181** and **185** on the two-dimensional material layer **150** so that the first and second upper electrodes **181** and **185** respectively correspond to the first and second electrodes **121** and **125**.

(94) As described above, in addition to the first and second electrodes **121** and **125**, when the first and second upper electrodes **181** and **185** are added on the two-dimensional material layer **150** to correspond to the first and second electrodes **121** and **125**, a double-sided contact structure with the two-dimensional material layer **150** is implemented, which may increase a charge injection path, and thus the contact resistance may be further lowered.

(95) FIG. 10 is a cross-sectional view schematically showing the electronic device **300** according to an embodiment, and shows an example implemented as a field effect transistor.

(96) Referring to FIG. 10, the electronic device **300** according to an embodiment may include the first and second electrodes **121** and **125** formed on the substrate **110** to be spaced from each other, a planarization layer **115** formed on the substrate **110** to bury the first and second electrodes **121** and **125** so that upper surfaces of the first and second electrodes **121** and **125** are exposed, the insertion layers **131** and **135** respectively formed on the first and second electrodes **121** and **125**, and the two-dimensional material layer **150** provided flatly over the first and second electrodes **121** and **125** in which the insertion layers **131** and **135** are respectively formed and the planarization layer **115**. The same reference numeral as in FIG. 5 denotes substantially the same configuration, and thus a repetitive description thereof is omitted herein.

(97) The planarization layer **115** may include an insulating material, for example, silicon oxide. As shown in FIG. 10, the two-dimensional material layer **150** may be formed in a structure in bottom contact with the first and second electrodes **121** and **125** and without additional strain application, by applying the planarization layer **115** burying the first and second electrodes **121** and **125** so that the upper surfaces of the first and second electrodes **121** and **125** are exposed, and forming flatly

the two-dimensional material layer **150** over the first and second electrodes **121** and **125** in which the insertion layers **131** and **135** are respectively formed and the planarization layer **115** by a transfer process.

(98) As shown in FIG. **10**, when the planarization layer **115** burying the first and second electrodes **121** and **125** so that the upper surfaces of the first and second electrodes **121** and **125** are exposed is applied, the two-dimensional material layer **150** may be formed in the structure in bottom contact with the first and second electrodes **121** and **125**, even if a two-dimensional material that is easily broken by strain during the transfer process is applied as the two-dimensional material layer **150**.

(99) As shown in FIG. **10**, when the planarization layer **115** burying the first and second electrodes **121** and **125** so that the upper surfaces of the first and second electrodes **121** and **125** are exposed is applied, the two-dimensional material layer **150** may be formed in a flat bottom contact structure.

(100) A structure in which the planarization layer **115** burying the first and second electrodes **121** and **125** so that the upper surfaces of the first and second electrodes **121** and **125** are exposed is applied may be formed in the following manner according to materials forming the first and second electrodes **121** and **125**.

(101) For example, when a metal capable of a planarization process, for example, a CMP process, such as a slurry, is used, in a state in which the planarization layer **115** is formed on the substrate **110** and a trench is previously deeply formed in a region where the first and second electrodes **121** and **125** is to be formed, a thick metal is deposited on the entire surface of the trench and then planarized by the planarization process, for example, the CMP process, and thus the structure in which the planarization layer **115** burying the first and second electrodes **121** and **125** is applied so that the upper surfaces of the first and second electrodes **121** and **125** are exposed without an air gap may be formed.

(102) As another example, when a metal incapable of the planarization process, for example, the CMP process, such as a noble metal, is used, the planarization layer **115** is formed on the substrate **110**, and a trench is shallowly formed to have various inclinations in a region in which the first and second electrodes **121** and **125** is to be formed, and thus the structure in which the planarization layer **115** burying the first and second electrodes **121** and **125** so that the upper surfaces of the first and second electrodes **121** and **125** are exposed is applied may be formed through a lift-off process in which the metal is deposited only up to the height of the trench.

(103) The two-dimensional material layer **150** may be formed in a flat bottom contact structure, by forming the insertion layers **131** and **135** on the first and second electrodes **121** and **125** of the planarized structure as formed above, and transferring the two-dimensional material layer **150** on the planarized structure.

(104) Even when the two-dimensional material layer **150** is provided over the first and second electrodes **121** and **125** and the planarization layer **115** as in FIG. **10**, since the two-dimensional material layer **150** is formed in the structure in bottom contact with the first and second electrodes **121** and **125**, a junction of a metal-two-dimensional material (semiconductor) without chemical bonding may be formed by transferring the two-dimensional material layer **150** onto the first and second electrodes **121** and **125**, and accordingly, the Fermi level pinning phenomenon may be reduced and/or minimized. In addition, a barrier may be lowered by controlling a work function, by providing the insertion layers **131** and **135** between the two-dimensional material layer **150** and the first and second electrodes **121** and **125**, and a contact resistance may be lowered by limiting and/or preventing oxidation of the first and second electrodes **121** and **125**.

(105) Therefore, even when the two-dimensional material layer **150** is formed in the flat bottom contact structure, a contact barrier, for example, a Schottky barrier, at the junction of the metal-two-dimensional material (semiconductor) may be lowered.

(106) FIG. **11** is a cross-sectional view schematically showing the electronic device **400** according to an embodiment, and shows an example implemented as a field effect transistor. The electronic device **400** shown in FIG. **11** is different from FIG. **10** in that the electronic device **400** further

includes the first and second upper electrodes **181** and **185** on the two-dimensional material layer **150** so that the first and second upper electrodes **181** and **185** respectively correspond to the first and second electrodes **121** and **125**. As described above, in addition to the first and second electrodes **121** and **125**, when the first and second upper electrodes **181** and **185** are added on the two-dimensional material layer **150** to correspond to the first and second electrodes **121** and **125**, a double-sided contact structure with the two-dimensional material layer **150** is implemented, which may increase a charge injection path, and thus the contact resistance may be further lowered.

(107) While the electronic devices **100**, **200**, **300**, and **400** described above with reference to FIGS. **5**, **9**, **10**, and **11** have been described as examples where the first and second electrodes **121** and **125** have a same size, example embodiments are not limited thereto.

(108) For example, FIG. **12** shows an electronic device **500** that is the same as the electronic device **100** in FIG. **5**, except the height **H1** of the first electrode **121** may be different than the height **H2** of the second electrode **125**. The height **H1** of the first electrode **121** may be less than the height **H2** of the second electrode **125**. Due to the different heights **H1** and **H2** of the first electrode **121** and the second electrode **125**, the strain of the inclined second portion **155** of the graphene **150** near the first electrode **121** may be different than the strain of the inclined portion **155'** near the second electrode **125**. An energy barrier of the graphene **150** on the first electrode **121** may be adjusted differently than an energy of the graphene **150** on the second electrode **125** by applying different strains to the inclined second portion **155** and inclined portion **155'** of the graphene **150**.

(109) While some embodiments in which electronic devices **100**, **200**, **300**, **400**, and **500** having the structure in which the two-dimensional material layer **150** is in bottom contact with a metal layer are implemented as field effect transistors have been described and illustrated above, the presented embodiments are merely examples and inventive concepts are not limited thereto.

(110) The electronic devices **1**, **10**, **100**, **200**, **300**, **400**, and **500** according to example embodiments may be implemented as various devices that may be required to lower a contact barrier by forming the two-dimensional material layer **150** in a structure in bottom contact with the metal layer. While embodiments have been described above with reference to the drawings, the embodiments are merely examples, and various modifications to the embodiments may be possible.

(111) In electronic devices according to example embodiments, a contact barrier, for example, a Schottky barrier, at a junction of a metal-two-dimensional material (semiconductor) may be lowered, by forming a two-dimensional material layer in a structure in bottom contact with a metal. In addition, in electronic devices according to example embodiments, by providing an insertion layer between a metal layer (an electrode) and the two-dimensional material layer, a barrier may be lowered by controlling a work function, and a contact resistance may be lowered by limiting and/or preventing oxidation of the metal layer.

(112) FIG. **13** is a circuit diagram of an electronic device according to an example embodiment.

(113) Referring to FIG. **13**, an electronic circuit **600** according to an embodiment may include a first transistor **Tr1** and a second transistor **Tr2** between a power terminal **Vdd** and a ground terminal. An input line may be configured to apply a voltage to the gates of the first transistor **Tr1** and a second transistor **Tr2**. An output line may be connected to a node between the first transistor **Tr1** and the second transistor **Tr2**. The electronic circuit **600** may include one of the electronic devices **100**, **200**, **300**, **400**, and **500** described above as the first transistor **Tr1** or second transistor **Tr2**.

(114) It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

Claims

1. An electronic device comprising: a substrate; a metal layer on a partial region of the substrate; a two-dimensional material layer over the metal layer and an upper surface of the substrate; and an insertion layer between the metal layer and the two-dimensional material layer, wherein the two-dimensional material layer directly contacts a first surface of the insertion layer, a second surface of the insertion layer directly contacts the metal layer, the second surface of the insertion layer is opposite the first surface of the insertion layer, and the two-dimensional material layer extends over a region of the upper surface of the substrate that is not covered by the metal layer.
2. The electronic device of claim 1, wherein the two-dimensional material layer comprises a transferable two-dimensional material.
3. The electronic device of claim 1, wherein the two-dimensional material layer comprises at least one of graphene, black phosphorus, or transition metal dichalcogenide (TMD).
4. The electronic device of claim 1, wherein the metal layer comprises a metal having a low work function or a noble metal.
5. The electronic device of claim 4, wherein the insertion layer comprises graphene, a-C, h-BN, or a-BN.
6. The electronic device of claim 1, wherein the insertion layer comprises graphene, a-C, h-BN, or a-BN.
7. The electronic device of claim 1, wherein the metal layer has a step with respect to a reference surface, and wherein the two-dimensional material layer comprises a first portion along the reference surface, a second portion on the metal layer; and an inclined third portion between the first portion and the second portion.
8. The electronic device of claim 7, wherein the two-dimensional material layer comprises a transferred two-dimensional material layer, and wherein the inclined third portion corresponds to a suspended portion.
9. The electronic device of claim 7, further comprising: an upper metal layer on the two-dimensional material layer, wherein the upper metal layer corresponds to the metal layer.
10. The electronic device of claim 1, further comprising: a planarization layer on the substrate, the planarization layer exposing an upper surface of the metal layer, wherein the two-dimensional material layer extends over the metal layer and the planarization layer.
11. The electronic device of claim 10, further comprising: an upper metal layer on the two-dimensional material layer, wherein the upper metal layer corresponds to the metal layer.
12. The electronic device of claim 1, wherein the metal layer comprises a first electrode and a second electrode spaced apart from each other on an upper surface of the substrate, wherein the two-dimensional material layer contacts an upper surface of the first electrode, two-dimensional material layer contacts an upper surface of the second electrode, and covers an upper surface of a part of the substrate between the first electrode and the second electrode, the part of the substrate between the first electrode and the second electrode is the region of the substrate not covered by the metal layer.
13. The electronic device of claim 12, further comprising: a gate insulating layer on the two-dimensional layer; and a gate electrode on the gate insulating layer, wherein the two-dimensional material layer is a channel layer.
14. The electronic device of claim 13, wherein the gate insulating layer comprises at least one of a high-k oxide, a silicon oxide, or a two-dimensional insulating material capable of atomic layer deposition (ALD).
15. An electronic device comprising: a substrate; a metal layer on a partial region of the substrate; a two-dimensional material layer over the metal layer and an upper surface of the substrate; and an insertion layer between the metal layer and the two-dimensional material layer, wherein the metal

layer comprises a first electrode and a second electrode spaced apart from each other on the substrate, and wherein the two-dimensional material layer covers an upper surface of the first electrode, an upper surface of the second electrode, and an upper surface of a part of the substrate between the first electrode and the second electrode.

16. The electronic device of claim 15, further comprising: a gate insulating layer on the two-dimensional material layer; and a gate electrode on the gate insulating layer, wherein the two-dimensional material layer is a channel layer.

17. The electronic device of claim 16, wherein the gate insulating layer comprises at least one of a high-k oxide, a silicon oxide, or a two-dimensional insulating material capable of atomic layer deposition (ALD).

18. The electronic device of claim 15, the two-dimensional layer extends over a region of the upper surface of the substrate that is not covered by the metal layer, the part of the substrate between the first electrode and the second electrode is the region of the substrate not covered by the metal layer, the two-dimensional material layer contacts the upper surface of the first electrode, the two-dimensional material layer contacts the upper surface of the second electrode, and the upper surface of the first electrode and the upper surface of the second electrode are parallel to each other.
